

TUSHAR NAGESH DESHMUKH

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Education

VIT Bhopal University

B.Tech in Computer Science and Engineering

2022 – 2026

CGPA: 8.55

Skills

- **Programming Languages:** Java, MySQL
- **Technical Skills:** DSA, Data Analytics, Web Development (HTML, CSS, Javascript), Cloud Computing (AWS)
- **Tools & Visualization:** Power BI, Exploratory Data Analytics (EDA), DAX

Projects

AI Animation Generator using Gemini and Manim

Aug 2025

LLM Application Project

- Developed an AI-powered text-to-animation system that converts natural language prompts into executable Manim animations using Google's Gemini LLM and LangChain.
- Engineered structured prompt templates and Pydantic-based output validation to generate reliable, error-free Python animation scripts.
- Automated the end-to-end workflow including code generation, script execution, video rendering, and playback within Google Colab.
- Implemented layout and rendering constraints to ensure visually consistent, non-overlapping animations while adhering to Manim best practices.

Pneumonia Detection Web Application — Chest Insight Project

Apr 2024 – May 2024

Healthcare AI Project

- Created a convolutional neural network (CNN) model using pre-trained VGG16 architecture on 1000+ chest X-ray images
- Developed a Streamlit-based web app that allows users to upload X-rays and receive real-time pneumonia detection results
- Achieved classification accuracy of 90%+ on validation datasets, demonstrating strong diagnostic potential
- Designed a user-friendly interface for doctors and patients, enabling seamless image upload and presenting a definitive pneumonia detection result.

Customer Churn Prediction — Machine Learning, Streamlit

Jan 2024

Predictive Modeling Project

- Performed data processing, cleaning, and feature engineering using pandas and NumPy on 5000+ customer datasets
- Trained and evaluated multiple machine learning models including Logistic Regression, SVM and Random Forest using scikit-learn; Logistic Regression achieved the best prediction accuracy of 89.5%
- Built interactive churn insights visualizations with matplotlib and reduced prediction latency for faster decision-making
- Developed and deployed a Streamlit-based web application improving business planning speed through real-time insights

Certifications

- AWS Certified Cloud Practitioner (CLF-C02)
- Coding Ninjas – Certificate of Excellence in Data Structures and Algorithms in Java

Leadership / Achievements

- Completed 100 Days of Code on LeetCode, solving problems in DSA using Java, also shared progress consistently on LinkedIn.
- AWS Cloud Club VIT Bhopal Core Member for the year 2023-24
- Managed resources for 10+ club events with 100+ participants each